# A. O. M. Shamsuddoha

Lecturer and Program Coordinator, Department of Electrical and Electronic Engineering
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## Profile

University lecturer in Electrical and Electronic Engineering with nearly six years of teaching experience and eight peer-reviewed publications, serving as Program Coordinator of the B.Sc. in EEE program at ULAB since 2025. Hands-on expertise in Cadence Virtuoso, Verilog and PSpice. Seeking a PhD in Electrical and Computer Engineering.

## Education

**M.Sc. in Electrical and Electronic Engineering** Coursework completed
*Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh*
CGPA: 3.50 / 4.00 (80%–89% marks)

**B.Sc. in Electrical and Electronic Engineering** December 2018
*Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh*
CGPA: 3.75 / 4.00 (75%–79% marks)
Thesis: *Low Power VLSI Circuit (SRAM) Design Using FinFET*. Supervisor: Prof. Satyendra Nath Biswas, PhD

## Research Experience

### Journal Articles

[J1] S. H. M. Atul, M. I. Hasan, S. E. Arafat, M. S. Islam, M. Hasan, A. O. Alfatha, F. Sultana and **A. O. M. Shamsuddoha**, "PHTF-Select: Phenology Aware Hurdle Foundation Modelling for Uncertainty Guided Crop Screening," *Natural Resources for Human Health*, vol. 6, no. 19S, pp. 1106–1131, 2026. doi: 10.53365/nrfhh.4052

[J2] M. Sakib, G. Hossain, T. Mahmud, A. Al Hasan, M. Nafis, M. Y. Hossain, N. M. Chisty, M. M. Mahmud, **A. O. M. Shamsuddoha** and M. M. Hossain, "SleepingEdge: Real-Time Sleep-Stage-Driven Epileptic Seizure Detection Using HMM and Hybrid OptiNet-SVM," *IEEE Access*, vol. 14, pp. 57646–57673, 2026. doi: 10.1109/ACCESS.2026.3682946

### Conference Papers

[C1] S. H. M. Atul, R. A. Shium, **A. O. M. Shamsuddoha**, M. Islam, M. A. Hossain and M. Hasan, "An Efficient and Interpretable Hybrid FCM-ELM Framework with Entropy-Guided Feature Selection for Multiclass Brain Tumor MRI Segmentation and Classification," in *2026 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2026. Accepted and presented.

[C2] M. M. L. Chowdhury, S. Sarder, M. R. Hossain, M. R. A. Panto, N. A. Mim and **A. O. M. Shamsuddoha**, "A Low-Cost IoT-Based Smart Home Automation and Security System with Energy-Efficient Operation for Sustainable Living in Developing Regions," in *2026 2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, Dhaka, Bangladesh, 2026. Accepted and presented.

[C3] S. H. M. Atul, **A. O. M. Shamsuddoha**, S. M. M. Islam, M. R. Biswas, H. K. F. Hai and S. M. Nawshed, "Detection and Classification of Poultry Diseases Using Deep Convolutional Neural Networks: A Ten-Class Image-Based Approach," in *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, 2025, pp. 462–467. doi: 10.1109/RAAICON69033.2025.11502034

[C4] S. Chakraborty, S. Zahir, N. T. Orchi, M. F. Bin Hafiz, **A. O. M. Shamsuddoha** and S. M. Dipto, "Violence Detection: A Multi-Model Approach Towards Automated Video Surveillance and Public Safety," in *2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, Dhaka, Bangladesh, 2024, pp. 1–6. doi: 10.1109/iCACCESS61735.2024.10499466

[C5] S. J. Hamim, M. K. Islam, **A. O. M. Shamsuddoha**, S. Dhar and A. K. M. Baki, "Performance Evaluation of $2 \times 2$ and Massive $8 \times 8$ MIMO at 28 GHz and 60 GHz for Dhaka City," in *2022 IEEE 7th International Conference for Convergence in Technology (I2CT)*, 2022, pp. 1–5. doi: 10.1109/I2CT54291.2022.9825246

[C6] **A. O. M. Shamsuddoha**, M. K. Islam and S. N. Biswas, "Design and Performance Analysis of SRAM Circuit Using Adiabatic Logic with FinFET," in *2021 International Conference on Electronics, Communications and Information Technology (ICECIT)*, Bangladesh, 2021, pp. 1–4. doi: 10.1109/ICECIT54077.2021.9641167

## Teaching Experience

**University of Liberal Arts Bangladesh (ULAB), Dhaka**
*Department of Electrical and Electronic Engineering*

**Program Coordinator, B.Sc. in EEE Program** February 2025 – Present

- Plan course offerings and class routines, and allocate teaching assistants
- Form capstone groups, organize exam question moderation, and prepare ISO and SDG reports
- Coordinate departmental academic meetings and continuous quality improvement (CQI) sessions
- Serve as the department's central point of contact for academic matters, liaising with university offices and other departments

**Lecturer** June 2023 – Present
*Courses: Electrical Circuits I; Electronic Circuits I; Electromagnetic Fields and Waves; Solid State Devices; Power Electronics; VLSI I; Microprocessor and Embedded Systems*

**University of Scholars, Dhaka**
*Department of Electrical and Electronic Engineering*

**Lecturer** April 2022 – May 2023
*Courses: VLSI I; Microprocessor and Interfacing; Electrical Properties of Materials; Electronic Devices and Circuits; Measurements and Instrumentation. Supervised several undergraduate projects in these areas.*

**Ahsanullah University of Science and Technology (AUST), Dhaka**
*Department of Electrical and Electronic Engineering*

**Part-Time Lecturer** December 2021 – May 2022
*Courses: VLSI I Lab; Microprocessor, Interfacing and System Design Lab; Energy Conversion II Lab*

**Daffodil International University, Dhaka**
*Department of Computer Science and Engineering*

**Lecturer** May 2019 – April 2020
*Courses: Electronic Devices and Circuits; Basic Electrical Circuits*

## Academic Service and Leadership

### Technical Reviewer

- International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka — October 2026
- International Conference on Computer and Information Technology (ICCIT), Cox's Bazar — December 2025
- International Conference on Computer and Information Technology (ICCIT), Cox's Bazar — December 2024
- International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka — March 2024

### Workshops Conducted

- Resource person, "Digital VLSI Circuit Design Essentials: Unleashing Creativity with Cadence Virtuoso Tool," Department of EEE, ULAB, with ULAB Electronics and Robotics Club and IEEE RAS ULAB Student Branch Chapter — March 2024

### Conference Presentations

- Poster presentation, International Conference on Electronics and Informatics 2022, Bangladesh Electronics and Informatics Society, representing University of Scholars — January 2023

### Organizing and Advising

- Advisor, ULAB Electronics and Robotics Club, University of Liberal Arts Bangladesh — May 2024 – July 2025
- Organizer, International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka — March 2024
- Advisor, Robotics Club, University of Scholars — November 2022 – May 2023
- Assistant Coordinator, MATLAB & Image Processing Competition, Engenius 2018 (inter-university tech competition), AUST — February 2018

## Professional Development

- Induction Training for Tertiary Teachers (24 hours, four modules), Center for Excellence in Teaching and Learning (CETL), ULAB — Fall 2025
- Two-Day Training on Outcome-Based Education (OBE), BAETE — May 2025
- Certificate Course on Teaching for Active Learning, Foundation for Learning Teaching and Research — September 2023
- Short course on Planning and Design of Electrical Substation, Continuing Education Center, University of Scholars — January 2023
- Training on Outcome-Based Curriculum and Accreditation Standards and Criteria, The International University of Scholars — April 2022
- Virtual BCI & Neurotechnology Spring School, g.tec medical engineering GmbH, Austria — April 2021
- CCNAv7 (Cisco Certified Network Associate, version 7), American International University-Bangladesh — March – November 2020
- Workshop on Introduction to Basic TCAD, United International University — August 2017
- Training on Web Programming Using PHP/MySQL, Institute of Information Technology, University of Dhaka — May – July 2016

## Honors and Awards

- 1st Runner-up, ROBORACE (Line Following Robot), Engenius 2018 inter-university tech competition, AUST EEE Society 2018
- 1st Runner-up, Tech Idea Race 2016 (innovative idea writing competition), IEEE AUST Student Branch 2016
- Perfect Attendance Award, Notre Dame College, for over 98% class attendance 2014

## Technical Skills and Languages

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|---|---|
| **VLSI design** | Cadence Virtuoso, Microwind |
| **Circuit simulation** | Cadence (OrCAD) PSpice, Proteus, Quartus II |
| **Programming and HDL** | Verilog, MATLAB, C++, Python |
| **Design tools** | AutoCAD, Tinkercad |
| **Languages** | Bengali (native), English (IELTS Academic 8.0, 2024), Hindi (basic), German (basic), Arabic (basic) |

## Projects and Competitions

- Performance evaluation of $2 \times 2$ and massive $8 \times 8$ MIMO at two frequencies for Dhaka City, *Advanced Telecommunication Engineering* March 2021
- Microcontroller-based DC to AC converter, *Power Electronics Lab* July 2018
- Digital door lock system in Proteus, *Digital Electronics II Lab* February 2018
- 4-bit up/down synchronous counter in Cadence, *VLSI I Lab* January 2018
- Participant, IEEEXtreme 11.0 Programming Competition October 2017
- Smoke detection system with protection, *Measurement and Instrumentation Lab* June 2016
- Home automation with GPS, *EEE Fest 2016 (intra-university project competition)* April 2016
- Participant, 11th and 12th Bangladesh Mathematical Olympiad (2013, 2014) and 2nd Bangladesh Physics Olympiad (2013)

Also took part in programming contests, innovative idea competitions, engineering olympiads and workshops.

## Professional Memberships

- IEEE, Student Member December 2016 – December 2025

  Societies: Computer Society (2023–2024); Power and Energy Society (2017–2018); Women in Engineering (2016–2018). Branch: IEEE AUST Student Branch (2016–2018).

## References

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